

Consumption of red maple in anticipation of beech mast-seeding drives reproduction in eastern chipmunks

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Abstract

1. Understanding the determinants of reproduction is a central question in evolutionary ecology. In pulsed-resource environments, the reproduction and population dynamics of seed consumers are driven by pulsed production of seeds by trees or mast-seeding. In Southern Québec, eastern chipmunks *Tamias striatus* exclusively reproduce during the summer before and the spring after a mast-seeding event of American beech. They thus seem to anticipate beech mast by reproducing during early summer, so that juveniles can emerge at the time of maximum beechnut abundance during late summer.
2. However, the cues allowing chipmunks to anticipate beech mast remain unknown, and the existence of the anticipation process itself has been questioned. To tackle those issues, we investigated the links between the nutritional ecology and reproduction of adult chipmunks and compared their spring diet in mast- versus post-mast years.
3. We monitored female reproductive status ($N = 446$), analysed cheek pouch contents at capture ($n = 3,761$ captures) and recorded seed production by deciduous trees on three different sites in Mont-Sutton from 2006 to 2018.
4. Results revealed a systematic shift in chipmunk diet towards red maple seeds in springs preceding a beech mast, with red maple seeds composing more than 77% of chipmunk diet. However, red maple consumption was unrelated to red maple production, but was related to beech seed production in the upcoming fall. We also found that red maple consumption best predicted the proportion of females in summer oestrus.
5. Our results confirm that chipmunks anticipate beech mast-seeding and highlight a key role of red maple consumption in that anticipation. Results also suggest that red maple seeds may contain nutrients or secondary plant components essential to sustain or trigger the summer reproduction in chipmunks, which allow them to remain synchronized with pulsed productions of both red maple and beech and improve their fitness.

KEYWORDS

Acer rubrum, *Fagus grandifolia*, fitness, food-hoarding, masting, reproductive success

1 | INTRODUCTION

Many plant species produce large seed crops at irregular intervals (Bogdziewicz et al., 2017; Koenig & Knops, 1998; Koenig, Mumme, Carmen, & Stanback, 1994; Norton & Kelly, 1988; Piovesan & Adams, 2001). Large seed productions by a population of plants or by several populations of a same or different species are typically referred to as masting or mast-seeding events (Kelly, 1994; Koenig & Knops, 1998; Koenig et al., 1994; Norton & Kelly, 1988). Mast-seeding is considered as an evolved strategy that confers fitness benefits to trees by limiting predation (Koenig et al., 1994; Nilsson, 2018), maximize pollination (Bogdziewicz et al., 2017; Koenig et al., 1994) or increase seed dispersal (Zwolak, Bogdziewicz, Wróbel, & Crone, 2016). Different mechanisms may drive mast-seeding depending on the species. However, masts usually result from a combination between the endogenous resources of trees and exogenous factors (Bogdziewicz et al., 2017; Piovesan & Adams, 2001). Species-specific response to climatic patterns has been identified as key parameters of mast-seeding. For instance, a moist-cool-summer year followed by a drought-summer year triggers the mast in European and American beech, *Fagus sylvatica* and *Fagus grandifolia* (Bogdziewicz et al., 2017; Piovesan & Adams, 2001). The wide among-year fluctuations in seed production generate pulses of resources that drive reproduction, survival and population dynamics of seed-eating animals (Bergeron, Réale, Humphries, & Garant, 2011; Bogdziewicz, Zwolak, & Crone, 2015; Boutin et al., 2006; Ogawa, Mortelliti, Witham, & Hunter, 2017; Schnurr et al., 2002). Overall, more than 100 vertebrate species, including 55 mammals and 67 birds, have been reported to respond to masts, including predators of seed-eating animals (Bogdziewicz et al., 2015).

Seed-eating animals have also evolved strategies to cope with pulsed-resources system and maximize their fitness (Bergeron et al., 2011; Boutin et al., 2006; Haines et al., 2018; Vander Wall, 1990). These include food caching and hoarding (Vander Wall, 1990), diet switching (Bogdziewicz et al., 2015) and anticipation of mast-seeding or anticipative reproduction (Bergeron et al., 2011; Boutin et al., 2006). An increased reproduction in anticipation of a mast has been recorded in several species, including the North American red squirrel *Tamiasciurus hudsonicus* (Boutin et al., 2006), the European red squirrel *Sciurus vulgaris* (Boutin et al., 2006; Wauters et al., 2008), the edible dormouse *Glis glis* (Lebl, Kürbis, Bieber, & Ruf, 2010; Pilastro, Tavecchia, & Marin, 2003) and the eastern chipmunk *Tamias striatus* (Bergeron et al., 2011). Yet, the underlying processes associated with this anticipation of mast-seeding have been far less studied than the post-mast responses of animals and, as a result, their existence is still debated (Boutin, McAdam, & Humphries, 2013; Boutin et al., 2006; King, 2013; White, 2007, 2013a, 2013b). The increased reproduction of small mammals before a mast may simply reflect opportunistic events or may be driven by unidentified factors other than the upcoming mast (White, 2013a, 2013b). In addition, the underlying

mechanisms or cues enabling individuals to anticipate a mast, to remain synchronized with pulsed-resource production and to increase their reproductive fitness have not yet been identified (Bergeron et al., 2011; Boutin et al., 2006).

Here we examine whether summer reproduction of eastern chipmunks indeed occurs in anticipation of an autumn mast or only as a response to early spring weather or an increased spring availability of supplemental protein- or energy-rich foods (Selonen & Wistbacka, 2016; White, 2007). In Southern Québec, eastern chipmunks exclusively reproduce during the summer before and the spring after a beech mast (Figure 1; Bergeron et al., 2011). Chipmunks thus seem to anticipate the beech mast by reproducing during early summer, so that juveniles can emerge at the time of maximum beechnut abundance during late summer (Bergeron et al., 2011). We therefore investigate the links between tree seed production, eastern chipmunk feeding ecology and reproductive effort in early summer. We assess the existence of marked differences in the diet of adult chipmunks between mast and non-mast years (i.e. post-mast years), which could represent cues of their anticipative summer reproduction. We also investigate whether complex links between local weather and tree seed productions of American beech, red maple *Acer rubrum* and sugar maple *Acer saccharum* may drive chipmunk reproduction.

We test the hypothesis that summer reproduction does not result from a systematic diet shift in response to spring changes in food availability that are synchronized with beech seed production, but instead as an anticipation of this mast. Species with a fast life history (smaller body mass, larger litter size, short longevity) generally display a reproductive response rather than a diet-switching response to mast-seeding (Bogdziewicz et al., 2015). Therefore, we predict that the diet of chipmunks will not shift towards a specific and dominant food category (i.e. no systematic increase in the consumption of a given spring food) in springs preceding a fall beech mast. Instead, the initiation of summer reproduction preceding a beech mast would be triggered by other cues associated with the beech mast-seeding, such as weather conditions or ingestion of secondary plant components from occasional consumption of tree buds and flowers (Descamps, Boutin, Berteaux, McAdam, & Gaillard, 2008; Elliott, 1978). Alternatively, in presence of a diet shift before summer reproduction (i.e. increase in the consumption of a given spring food; see White, 2007), two alternative predictions can be made. First, that such food serves as a source of energy, proteins, vitamins and minerals triggering and sustaining reproduction (White, 2007, 2013b), and chipmunks do not anticipate the mast. In that case, the consumption of the spring food should be correlated with its production and both should best predict summer reproductive effort. Second, the spring food source only serves as a 'cue' of the upcoming beech mast, for example this food source contains a plant-derivative molecule (Berger, Negus, Sanders, & Gardner, 1981; DeGabriel, Moore, Foley, & Johnson, 2009) that triggers reproduction. In that case, the spring food production should be correlated with beech production in the

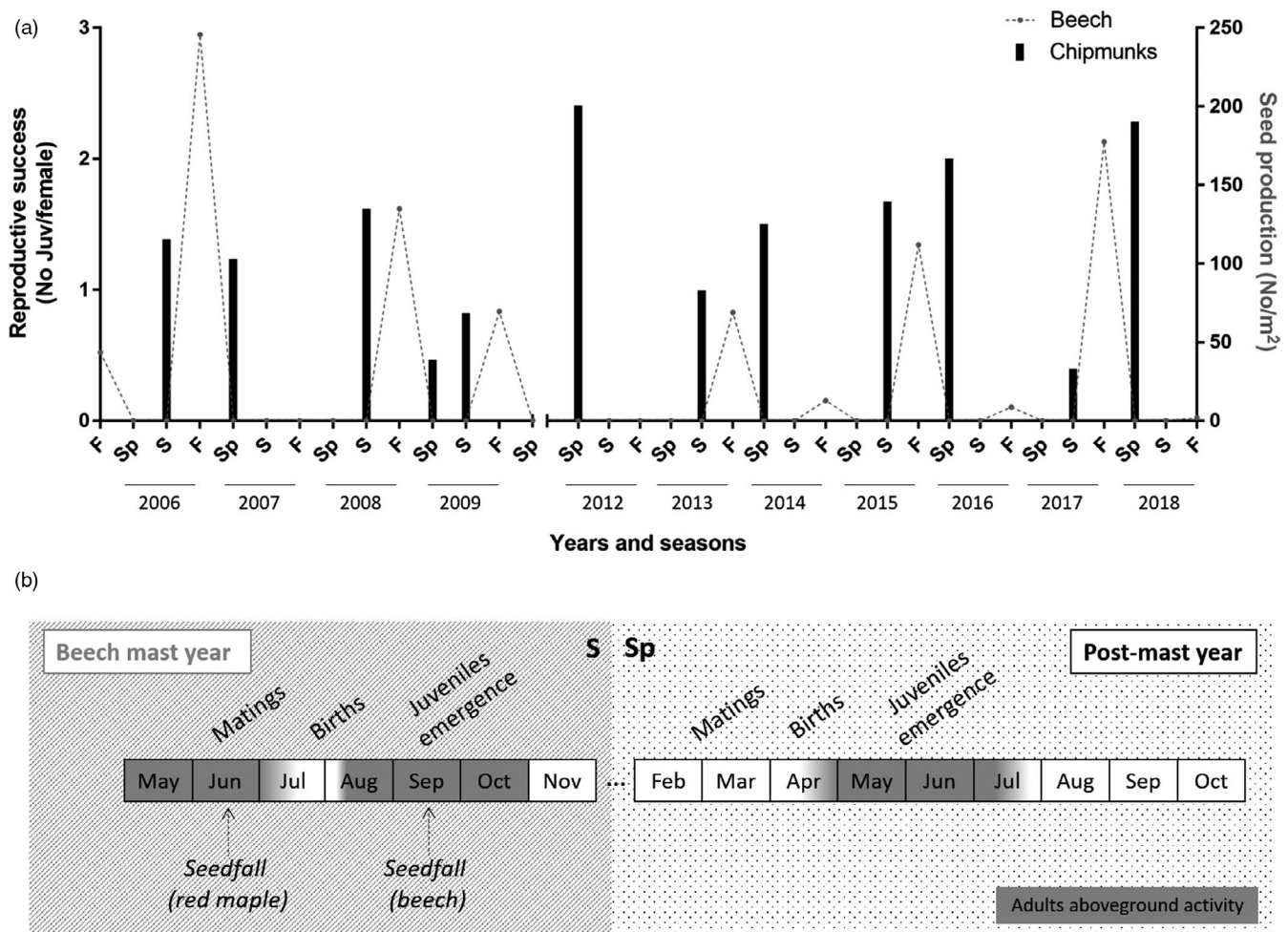


FIGURE 1 Beech mast-seeding and chipmunks reproduction. In (a) the reproductive success of eastern chipmunks *Tamias striatus* and beech *Fagus grandifolia* seed production are represented from 2006 to 2018. Sp, S and F represent Spring, Summer and Fall respectively. The black bars represent the average number of juveniles captured per adult female respectively before (summer) and after a beech mast (spring). The grey dots and lines represent the estimated production of beechnuts (number of seeds/m²) in the fall of each year. Data are represented from 2006 to 2009 (site 1) and from 2012 to 2018 (average values of sites 2 and 3). In (b) phenology of chipmunk reproduction preceding (summer reproduction-S) and following (spring reproduction-Sp) a beech mast. The above-ground activity period of adults is represented by a grey coloration (Bergeron et al., 2011)

fall, and should best predict summer reproductive success. Finally, we also investigate whether local weather, known to be related to reproduction of small mammals (Selonen, Wistbacka, & Korpimäki, 2016) may affect chipmunk reproduction at the population level.

2 | MATERIALS AND METHODS

2.1 | Model species, study site and data collection

Eastern chipmunks are omnivorous, food-storing sciurid rodents that display periodic torpor events in winter (Elliott, 1978; Humphries, Thomas, Hall, Speakman, & Kramer, 2002; Humphries, Thomas, & Kramer, 2003; Vander Wall, 1990). They feed on a variety of plants (bulbs, tree seeds, flowers, fruits, etc.), mushrooms and invertebrates throughout the season. They

also forage on non-perishable foods such as nuts and seeds late in the summer and in the fall to constitute their hoards (Elliott, 1978). Eastern chipmunks may display two reproductive seasons per year, when food availability is high (Smith & Smith, 1972). In southern Québec, however, they usually alternate a year with summer reproduction and a year with spring reproduction (Bergeron et al., 2011; see Figure 1). Juveniles born in summer (before a beech mast) can reproduce for the first time the following spring (at approximately 7 months), whereas juveniles born in spring (after a beech mast) usually reproduce for the first time during the next summer, at approximately 14 months (Bergeron et al., 2011; Montiglio et al., 2014). They rarely reproduce more than once per year (see Figure 1, with an exception in 2009). As food-storing animals, chipmunks may rely on their food hoards for up to 11 months when food availability is low (Elliott, 1978; Munro, Thomas, & Humphries, 2005; see Figure 1, year 2007). On the monitored sites, breeding always occurs in early March for

spring reproduction, whereas it occurs around the 15th of June for summer reproduction, with all reproducing females displaying signs of oestrus within 2 weeks (M.L. Tissier, pers. obs.). Such level of breeding synchronization at the site level, but also between sites and mast-seeding years, strongly indicates that there is an external cue leading to reproductive synchrony (Ims, 1990) and that chipmunk reproduction is not solely driven by intrinsic cues (such as body condition, found to influence timing of reproduction in fat-storing rodents; Fietz, Pflug, Schlund, & Tataruch, 2005). Thereby, we focus on external cues that may induce female oestrus before beech mast-seeding.

We monitored eastern chipmunks in a deciduous forest of southern Québec, Canada (45°05'N, 72°26'W), on one site from 2006 to 2010 (site 1, 25 ha) and two sites from 2012 to 2018 (sites 2 and 3, 6.8 ha each). All three sites were located in Mont Sutton, at <10 km of distance from each other. On those sites, American beech trees and sugar maples dominated canopy-forming trees (Landry-Cuerrier, Munro, Thomas, & Humphries, 2008) followed by red maples, whereas trout lilies *Erythronium americanum* and spring beauties *Claytonia caroliniana* were abundant herbaceous plants. Red maple seed production occurred in mid-June/early July, whereas beech trees and sugar maples produced their seeds in the late summer/early fall (by the end of September). Since 2005, beech mast-seeding occurred approximately every 2 years, with the exception of 2008–2009 when beech masts occurred 2 years in a row (Figure 1a). However, in 2009, there has been a 'false' beech mast as most beechnuts were empty, resulting in an absence of chipmunk reproduction the following spring (2010; see Bergeron et al., 2011).

Each year, from the end of April to October (site 1) and from May to September (sites 2 and 3), we conducted daily trapping using Longworth traps, from 8:00 a.m. until dusk (Bergeron et al., 2011; St-Hilaire, Réale, & Garant, 2017). Individuals were systematically trapped and marked with ear tags and a Trovan PIT tag (Bergeron et al., 2011). We recorded the body mass, sex, reproductive status and determined the age class (juveniles or adults) of each individual captured. Females with a swollen vulva, with largely developed mammae, or lactating, were considered as reproductively active. Individuals were classified as juveniles if they weighed <70 g at emergence/first capture or if they did not display any sign of sexual maturity over the entire season (i.e. absence of darkened scrotum for males and undeveloped mammae for females; Bergeron et al., 2011; Smith & Smith, 1972). We considered as adult individuals weighing >80 g at first capture in May or showing reproductive signs of sexual maturity (individuals weighing between 70 and 80 g at first capture and that could not be classified as juveniles or adults were considered as missing data). Juveniles first captured before 1st August were attributed to spring reproduction, whereas juveniles first captured after 1st August were attributed to summer reproduction (Bergeron et al., 2011). We captured 253 and 267 juveniles from the summer and spring reproductions, respectively, and 446 females with a known reproductive status during the 11 years of this study (Table S1).

2.2 | Mast-seeding, food pouches content and weather data

Seed productions of beech, red and sugar maples were quantified using 30 plastic buckets with a 29.1 cm diameter for site 1, and 13 plastic buckets with a 27.9 cm diameter for each of sites 2 and 3. Buckets were evenly distributed on the trapping grid and placed under the canopy of trees with a circumference >40 cm at breast height (Figure S1). Buckets were more than 40 cm deep, placed 1 m away from the trunk of the target tree, and mounted on two smooth metal posts at >50 cm above ground level (Bergeron et al., 2011). We collected bucket contents in June for red maples and in October–November for beech and sugar maple. For each tree species, we recorded the number of fruits (i.e. envelopes with an embryo, considering that for the beech, the envelope contains two embryos or more), seeds (embryos without an envelope) and empty envelopes. We only considered the fruits and seeds collected from the bucket (and excluded the empty envelopes) to estimate yearly tree production, as a number of seeds/m². Red maple seeds were not collected in 2005 and 2012 and there was no collection of sugar maple seeds in 2016.

Between 2006 and 2018, we captured 719 adult and 602 juvenile chipmunks (21,175 capture events). Chipmunks collect large amount of food in their cheek pouches that they release in the trap at capture (Elliott, 1978; Vander Wall, 1990). We recorded chipmunk cheek pouches content from the food expelled in the traps at each capture from 2006 to 2010 and from 2012 to 2018 (3,761 captures with food from 339 adults and 116 Juveniles). We counted and identified to the species the most common food items (beech, red maple, sugar maple, trout lily, spring beauty and black cherry *Prunus serotina*). Other food items, less abundant and not easily identifiable to the species, were counted and grouped into the following three categories: mushrooms, invertebrates (e.g. slugs, beetles, caterpillars, millipedes, snails, woodlice and undetermined insects) and 'other plants' (e.g. ironwood, fern, basswood, *Viburnum* sp. and raspberries).

We obtained weather data from Environment Canada (<https://weather.gc.ca/>) at the closest meteorological station in Sutton (Québec, Canada) located at approximately 20 km from our study sites. Information recorded at the station reflects local weather data of our sites (Dammhahn, Landry-Cuerrier, Réale, Garant, & Humphries, 2017). The data obtained include average minimum and maximum monthly temperatures (°C), total monthly rainfall (mm) and monthly total snowfall (cm). We used monthly values in June (period of summer oestrus), or average values between March and June.

2.3 | Data analyses

2.3.1 | Indexes of chipmunk diet composition

We analysed the diet of chipmunks at the population level by calculating the frequency of occurrences (FO_i) and the percent (%N_i) of each food item *i* in the pouches of adults at capture (adapted from Holíšová, 1977; Hyslop, 1980). FO_i varies from 0 (no individuals captured with item *i*) to 100 (all captured individuals expelled the item

i at each capture). It gives information on the part of the population consuming a food item and may thereby inform us on whether a diet shift occurs at the population level or only in some individuals:

$$FO_i = \frac{NC_i}{NC} \times \frac{NI_i}{NI} \times 100, \quad (1)$$

where NC_i is the number of captures with item i , NC is the total number of captures, NI_i is the number of individuals captured with item i and NI is the total number of individuals captured with at least one food item; $\%N_i$ provides quantitative information on the abundance of each item in the diet of chipmunks:

$$\%N_i = \frac{N_i}{N} \times 100, \quad (2)$$

where N_i is the number of items i collected and N is the total number of items collected.

We also calculated the relative quantity of each food item i (RQ_i). This index allows to quantitatively estimate the preponderance of each food category in the diet of the population, while avoiding the bias caused by overabundant items collected in large amounts by specialists:

$$RQ_i = FO_i \times \frac{N_i}{100}. \quad (3)$$

Annual RQ_i of the dominant food items will be used in subsequent analyses related to reproduction.

We used GLMs with a Tweedie distribution to investigate whether a diet shift occurred towards a specific item between post-mast and mast years (i.e. mast effect on FO_i and $\%N_i$, models 1 and 2). To evaluate if the diet shift was systematic, that is occurred each spring preceding a mast, we used a similar approach with the RQ_i , investigating potential inter-annual effects on the relative quantity of each food in the diet of the population (RQ_i , model 3). The Tweedie distribution is adapted to treat data with a right-skewed distribution and high proportions of exact 0s (Bonat & Kokonendji, 2017). We included beech mast (Yes/No, models 1 and 2) or years (model 3). For all models, we included season (spring or summer), identity of the food category and their interactions. We conducted pairwise multiple comparisons using the Bonferroni procedure. When investigating the effects of all those variables on FO_i , we found a significant three-way interaction between mast, season and food category (model 1; Table S3). The purpose of this study being to investigate the determinants of the summer reproduction, which is initiated in late spring, we will hereafter focus on the diet of adult chipmunks in spring (i.e. from first capture in late April until the end of June). However, information on the summer diet of chipmunks can be found on Figure S2.

2.3.2 | Tree seed production and relationship with chipmunk food consumption

We calculated the average production of fruits/seeds per site and per year, using a GLMM (Poisson distribution, log-link function). We

included year as a fixed effect and bucket identity as a random effect. We then used the estimated value (average number of seeds produced per site and per year) to calculate the yearly average number of seeds/m² for each site. We analysed the association between yearly maple and beech productions, using nonparametric spearman correlations. We then analysed the respective effects of red maple and beech seed productions on chipmunks' seed consumption (RQ_i of the dominant food item) using linear regressions.

2.3.3 | Environmental determinants of chipmunk summer reproduction

We tested which environmental factors were the most strongly related to the number of females in summer oestrus (captured with a swollen vulva) using a GLM. We considered this variable of reproduction as it best represents the initiation of a reproduction compared to the number of lactating females or the number of juveniles captured (see Supporting Information results for models on other variables of reproduction). Given that body mass is strongly related to the reproductive stage, especially in females (Elliott, 1978), we could not complement the analysis of the environmental determinants of reproduction by an analysis of body mass (see Fietz et al., 2005) because of an inherent redundancy in the analyses. Indeed, in mast years, females have not initiated reproduction when first captured in May whereas in post-mast years they are about to wean their juveniles at that time (Figure 1b).

This GLM was conducted using count data as response variables with a negative binomial distribution (log-link function). We included the log number of captured adult females with known reproductive status as an offset term in the model to account for an uneven number of observations between years. This analysis is thereby equivalent to an analysis of proportions. In each model, sites (1, 2 or 3), seasonal or monthly weather variables (rainfall, minimum and maximum average temperatures), tree seed production variables (as a yearly number of seeds/m²) and the spring RQ_i of the dominant food items highlighted by previous analyses were included as independent variables. We carried out a model selection based on the lowest AICc value (models with a $\Delta AICc > 6$ compared to the model with the lowest AICc were systematically excluded; Symonds & Moussalli, 2011). For each of the remaining models (with a $\Delta AICc < 6$), we calculated the evidence ratio (ER) and the Akaike weight (w_i) to assess the relative strength of each candidate model (Symonds & Moussalli, 2011). The ER provides a measure of how much more likely the model x is to be the best model than model y : a value of 5 when comparing model y to model x indicates that model x is 5 times more likely to be the best approximating model than model y . The Akaike weight w_i , comprised between 0 and 1, is equivalent to the probability that a given model is the best approximating model (Symonds & Moussalli, 2011).

Analyses were conducted using IBM SPSS software (IBM SPSS Statistics for WINDOWS v. 24.0. IBM Corp.). Figures were prepared using GraphPad Prism (Version 7.00. for WINDOWS).

3 | RESULTS

3.1 | Pulsed resources and chipmunks feeding ecology

3.1.1 | Beech mast-seeding effects on the FO_i and $\%N_i$ of each food item in chipmunks' diet

The FO_i , which informs on the part of the population consuming a given item, was significantly affected by the mast*season*food category interaction (Table S3). In springs preceding a summer reproduction and a beech mast, $FO_{red\ maple}$ was nine times higher than in post-mast years (mean difference = 23.46 ± 6.48 , Figure 2a, $p < .001$). In parallel, $FO_{spring\ beauty}$, $FO_{trout\ lily}$ and $FO_{sugar\ maple}$ (from the previous fall) were lower in mast than in post-mast years (mean differences: -8.30 ± 3.37 , $p = .014$; -25.37 ± 7.38 , $p = .011$; -2.80 ± 1.10 , $p = .001$ respectively; Figure 2a). We found no mast effects on spring FO_i of other food items (beech, mushrooms, invertebrates, cherry, other plants and unknown food; Figure 2a; $p > .05$).

We also found a significant interaction between mast and food category on the $\%N_i$ ($\chi^2 = 55.69$, $p < .001$), which informs on the

relative quantity of each item in the diet of the population. In addition to being consumed by a large part of the population, the abundance of red maple seeds in the diet of chipmunks was significantly higher in mast compared to post-mast years, as revealed by the fourfold increase in the spring $\%N_{red\ maple}$ (Figure 2b, mean difference = 61.75 ± 18.39 , $p < .001$). As a consequence, $\%N_{spring\ beauty}$, $\%N_{trout\ lily}$ and $\%N_{sugar\ maple}$ were lower in mast compared to post-mast years (mean differences = -12.05 ± 4.76 , $p = .011$; -39.31 ± 10.33 , $p < .001$; -2.73 ± 1.30 , $p = .035$ respectively; Figure 2b). We found no beech mast effect on spring $\%N_i$ of other food categories (Figure 2b; $p > .1$).

3.1.2 | Yearly changes in the relative quantity of each food item in the spring diet of chipmunks

Spring RQ_i significantly differed among years, but only for some food categories (year*food category interaction, $\chi^2 = 253.93$, $p < .001$, Figure 2c). This yearly difference in RQ_i was mostly due to a significant increase in $RQ_{red\ maples}$ in springs preceding a beech mast (Figure 2c) compared to post-mast years, except for 2018 ($p > .05$). Spring $RQ_{red\ maple}$ represented from 81.2% to 97.7% of total RQ_i in

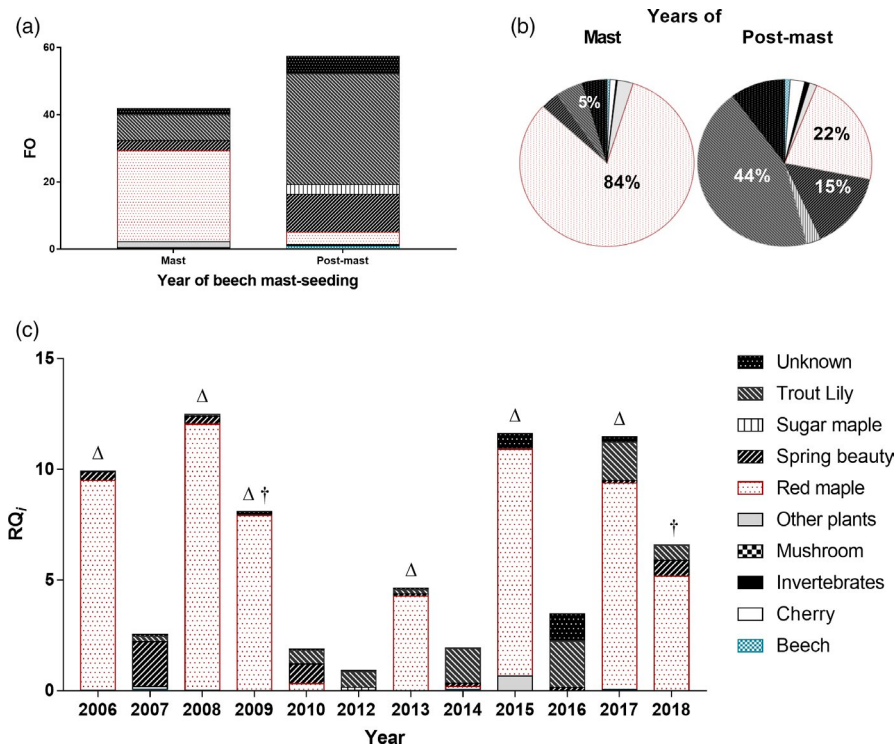


FIGURE 2 Diets of adult chipmunks in spring. In (a), frequency of occurrences (FO_i) of 10 food categories collected by adult chipmunks in springs of beech mast years and springs of post-mast years. Different letters highlight significant differences between the groups ($p < .05$). In (b), percent of 10 food categories ($\%N_i$) in beech-mast-year springs and post-mast years. The framed percentages represent the food categories for which the difference between mast- and post-mast years was significant ($p < .05$). In (c) yearly spring RQ_i of each food category. The Δ symbol highlights years of beech mast-seeding. The $\dagger$ symbol highlights two particular years (2009 and 2018) during which females initiated a summer oestrus while terminating spring reproduction (following a beech mast the previous fall; see Figure 1). This led to a summer reproduction in 2009 but not in 2018 (see below). The 'invertebrates' category includes slug, beetles, caterpillars, millipedes, snails, woodlouse and undetermined insects. The 'other plants' category includes ironwood, fern, basswood, *Viburnum* sp. and raspberries. Mast years: 2006, 2008, 2009, 2013, 2015 and 2017 and post-mast years: 2007, 2010, 2012, 2014, 2016 and 2018

most years, whereas it represented <6% of total RQ_i in post-mast years (with the exception of 2018, where RQ_{red maple} composed 77.8% of total RQ_i; see Figure 2c). However, we found no significant linear relation between red maple consumption by chipmunks (RQ_{red maple}) and red maple production in spring ($R^2 = .208$, $p = .076$; Figure 3a). Alternatively, we found that spring RQ_{red maple} was significantly and positively related to beech seed production in the upcoming fall ($R^2 = .704$, $p < .001$; Figure 3b). Results also show a positive effect of the interaction between beech seed production and red maple consumption on the proportion of females in oestrus (Figure 3b).

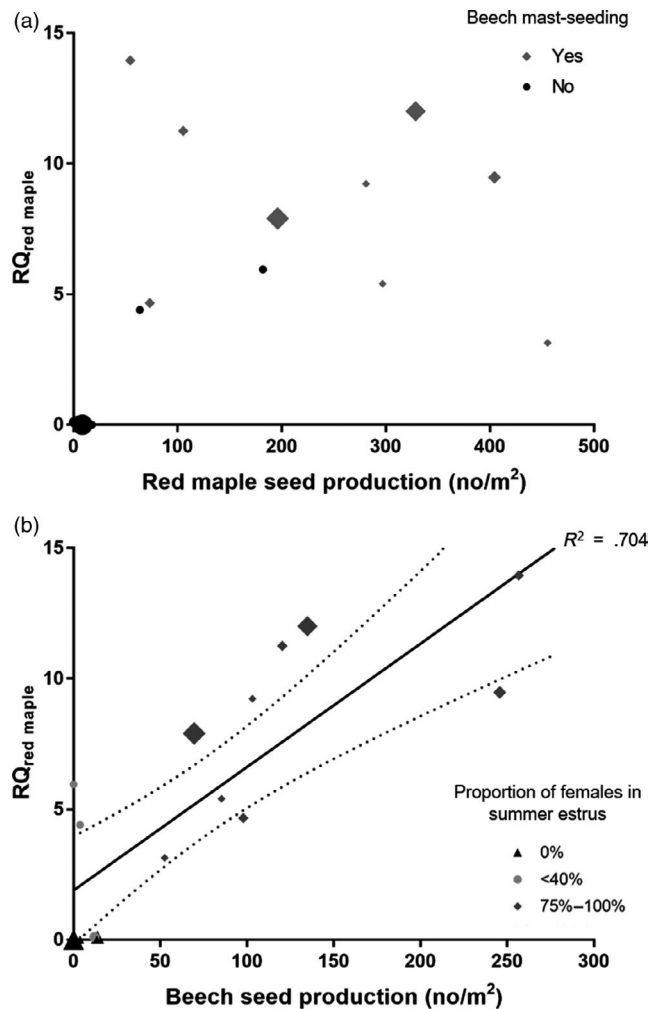


FIGURE 3 Effects of red maple and beech seed productions (no seeds/m²) on consumption of red maple by chipmunks in spring (relative quantity in the diet of the population; RQ_{red maple}). In (a), the relation between red maple production and spring consumption by chipmunks is represented ($R^2 = .208$, $p = .076$). Diamonds represent years of beech mast whereas years with no beech mast are represented by black circles. In (b), the relation between red maple consumption in spring and beech seed production in the upcoming fall is represented ($R^2 = .704$, $p < .001$). In (b), the proportion of females in oestrus in the population is highlighted. Each dot represents the yearly seed production of red maple or beech for each site. The size of the dots is scaled to the number of adult females with known reproductive status captured that year

3.2 | Mast-seeding, red maple consumption and chipmunk reproduction

3.2.1 | Mast-seeding of beech and maples

Beech mast-seeding occurred every 2 years on average (with an exception in 2008–2009; Figure 4). Red maple mast-seeding in spring generally preceded beech mast-seeding in the fall, with the exception of 2018 when a red maple mast-seeding occurred in the absence of a beech tree mast (Figure 4). Sugar maple mast-seeding events occurred less frequently and intensely than the two other species (Figure S3). Therefore, and given that sugar maples produce seeds in the fall and unlikely provide cues for chipmunk summer reproduction, we excluded it from our subsequent analyses. We found a positive and significant correlation between red maple and beech seed production (Spearman's Rho = 0.505, $p = .046$). Beech mast-seeding in the fall occurred only when red maple seed production was high (min average value of 54.56 seeds/m² and median value of 104.56; Figure S4).

3.2.2 | Environmental determinants of chipmunk reproduction

Red maple seed production and consumption, and beech seed production were all significantly and positively related to oestrus in female chipmunks (Figure 5).

Because of the strong relation between RQ_{red maple} and beech seed productions (Figure 3b), these two variables could not be included in the same models as it would create problems of collinearity. The best approximating model (i.e. model 1, with the lowest AICc, performing 2.34 to 14.66 times better than other candidate models and with a w_i of 0.53) excluded beech seed production and revealed a significant and positive effect of red maple consumption on the number of females initiating oestrus (Table 1 and Figure 5a; in Figure 5, the variable is represented in proportions) and a slight effect of red maple production (Table 1; see Table S6 for outputs of other candidate models). We found similar effects for other variables of reproduction, namely the number of lactating females and the average number of juveniles per female (see Tables S8 and S9; Figure S5). Other candidate models, including beech seed production instead of RQ_{red maple}, were respectively 5.10 to 14.66 less likely to be the best approximating model than model 1 and had a probability of being the best approximating model ≤ 0.1 (Table 1, models 3 and 5). Weather variables were all systematically excluded from the best approximating model but when conserved, they appeared in models with a $\Delta AICc > 4$ compared to the best approximating model, with an ER > 5, a $w_i < 0.1$ (e.g. model 4, Table 1) and their effects were non-significant (see Tables S6, S8 and S9), suggesting that they have little influence on chipmunks summer reproduction.

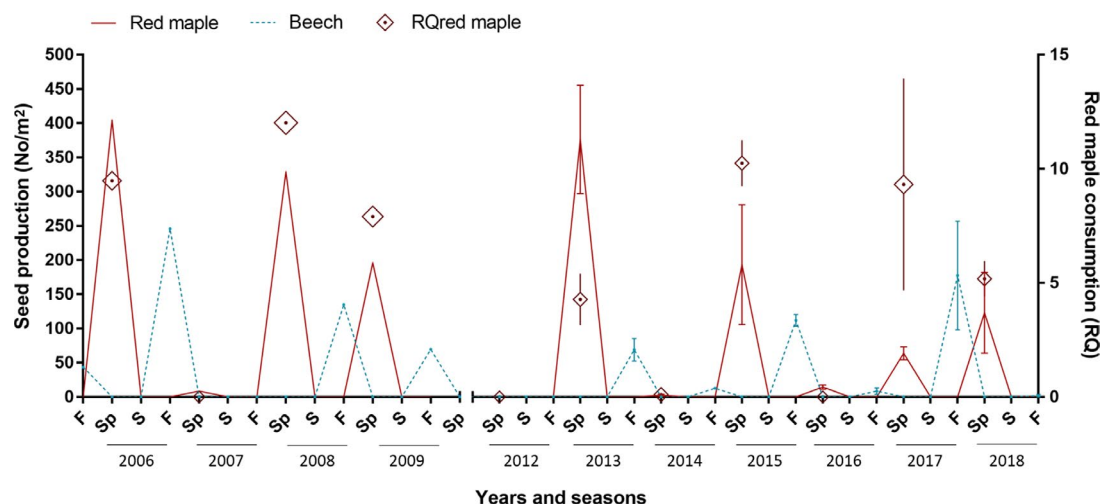


FIGURE 4 Productions of beech and red maple seeds with corresponding consumption of red maple by chipmunks per year and per season from 2006 to 2018. Beech and red maple average seed production (number of seeds/m²) are represented in fall for beech (blue dots and dashed lines, F) and spring for red maple (red lines, Sp). Spring consumption of red maple ($RQ_{red\ maple}$) is represented by a dark-red diamond with a dot. The size of the diamond is scaled to the number of adults captured with food in its pouches in spring. Data are represented from 2006 to 2009 (site 1) and from 2012 to 2018 (average values of sites 2 and 3; $M \pm SE$). Red maple production was not recorded in 2012

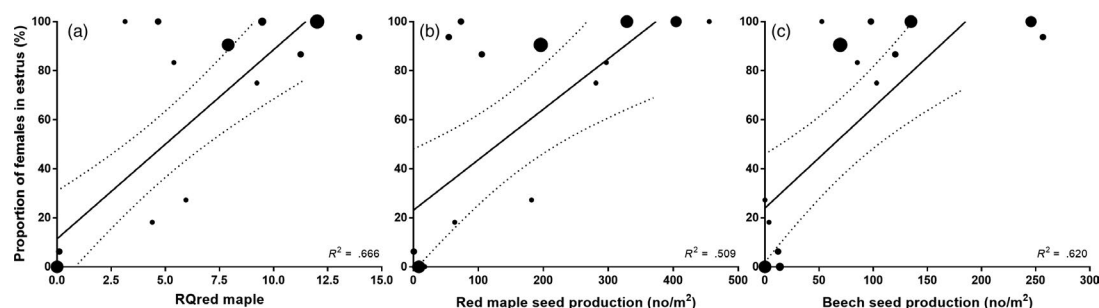


FIGURE 5 Proportion of chipmunk females in oestrus according to red maple consumption ($RQ_{red\ maple}$; a), red maple seed production (b) and beech seed production (c). The 95% confidence intervals are represented by dotted lines. Red maple consumption is expressed as a relative quantity of seeds in the diet of chipmunks in spring ($RQ_{red\ maple}$). Red maple and beech seed productions are expressed as a number of seeds/m² ($p < .05$). Each dot represents the grid-year combination (i.e. the yearly % of females in oestrus for each site). The size of the dots is scaled to the number of adult females with known reproductive status captured that year

TABLE 1 (A) Best-ranked regression models including the model with the lowest AICc and all models with $\Delta AICc < 6$ and (B) output of the best approximating model (model 1; GLM, negative binomial with an offset term) investigating the effects of environmental variables on the number of adult females initiating a summer oestrus in the population. In (A) the evidence ratio (ER) and the Akaike weight (w_i) are presented for each candidate model. RM, Red maple seed production (spring); BE, Beech seed production (fall); T_a June, average ambient temperature in June. In (B) The coefficient B is represented with standard residuals (SR). Outputs of models 2–5 are presented in Table S6B–E. The year 2012 is not included in this analysis since no information was available for red maple production (but see Table S7 for analyses including 2012)

A – Candidate models	AICc	ΔAICc	ER	wi	
1. RQ _{red maple} + RM	101.91			0.54	
2. RQ _{red maple}	103.56	1.70	2.34	0.23	
3. RM + BE	105.17	3.26	5.10	0.10	
4. RQ _{red maple} + T _a June	105.39	3.48	5.70	0.09	
5. BE	107.28	5.37	14.66	0.04	
B – Effects (model 1)	χ ²	df	B	SR	p value
Constant	4.906	1;13	−1.357	0.613	.027
RQ _{red maple} (Spring)	8.113	1;13	0.226	0.080	.004
Red maple production (Spring)	4.138	1;13	0.004	0.002	.042

4 | DISCUSSION

Contrary to our initial predictions, we found that chipmunks systematically displayed a diet shift in springs preceding a beech mast by increasing their consumption of red maple seeds much more than expected by the increase in red maple seed production. Indeed, red maple consumption in spring was not related to red maple production, but was instead strongly related to beech seed production in the upcoming fall. This suggests that some cues in the environment may lead chipmunks to increase their consumption of red maple and specialize on this food resource in anticipation of beech mast-seeding. Although we found that red maple and beech generally mast in the same years (except in 2018), we found no strong quantitative correlation in their productions of seeds. Red maple seed production is thus unlikely the cue that triggers chipmunk reproduction. Finally, rather than red maple or beech seed production, red maple consumption best predicted the number of females initiating a summer oestrus. Our results therefore highlight complex links between mast-seeding of beech and red maple, feeding ecology and reproduction of the eastern chipmunk at the population level.

Diet-switching responses to mast-seeding rarely occur in species with fast life history, namely with low body mass, large litter size and short life span (Bogdziewicz et al., 2015). However, the authors of this meta-analysis highlighted that most studies only used 1 or 2 years of data on animal diets, and thereby implicitly ignored the effects of pulsed resources associated with masts (Bogdziewicz et al., 2015). Using a long-term monitoring of chipmunks food collection and tree seed production, we report a systematic shift in the diet of a species with a fast life history (Bergeron et al., 2011; Smith & Smith, 1972). The four to ninefold increase in red maple seed consumption was solely recorded in springs preceding a summer reproduction and a beech mast. It was systematically associated with a decrease in the consumption of bulbs of trout lily and spring beauty and seeds of fall-producing trees such as sugar maple, highlighting a diet shift towards red maple. As a consequence, in springs preceding a summer reproduction and a beech mast, chipmunks would have access to twice more proteins and lipids (Table S5) and 25% less carbohydrates than in post-mast springs (see Supporting Information based on data from the literature in Tables S2, S4 and S5). However, red maple consumption was not related to red maple seed production, but to upcoming beech seed production. This suggests that the shift towards red maple occurred in anticipation of beech mast-seeding and not in response to red maple mast-seeding. In addition, we found that red maple consumption best predicted reproductive effort at the population level. Altogether, our results suggest that red maple consumption is both a cue of beech mast-seeding and also potentially a trigger (or at least a substrate) for summer reproduction.

Red maples flower in April–May in North-eastern America (Primack & McCall, 1986; Walters & Yawney, 1990), which corresponds to chipmunk emergence from hibernation (Elliott, 1978; Smith & Smith, 1972). Seeds generally ripen and fall to the ground in June (Elliott, 1978). This coincides with female 'summer oestrus' dates, occurring from June 13th to July 10th on our sites. Seeds of

red maples have been reported as a key food resource for eastern chipmunks (Elliott, 1978). Individuals were observed to cover very long distance to access red maple seeds, even though other food sources, such as trout lily, were still very abundant in the environment (Elliott, 1978). Red maple has great nutritional value for mammal reproduction, as its tissues are very rich in energy, proteins, minerals and essential vitamins (Bonner, 1974; Burkholder & McVeigh, 1944). Its seeds contain one of the highest contents in proteins among tree-produced seeds (i.e. above 20%) and high levels of magnesium (Bonner, 1974). In addition, the buds of red maples contain high levels of B-vitamins (Burkholder & McVeigh, 1944). In particular, it presents exceptionally high levels of niacin or vitamin B3 (i.e. 4–19 times more than other deciduous trees and conifers; Burkholder & McVeigh, 1944). Dietary content in niacin is crucial for mammal reproduction (Mishra, Aderao, Chaudhary, & Raje, 2018; Tissier, Handrich, Dallongeville, Robin, & Hahold, 2017; Wang et al., 2018). Deficiencies in this vitamin lead to reproductive failure in another food-hoarding rodent, whereas dietary supplementation in this vitamin led to above-average reproductive rates (Tissier et al., 2017). Results presented here therefore partly support the hypothesis that adults have access to an additive protein-rich food that sustains their reproduction (White, 2007, 2013a, 2013b). However, they also corroborate previous findings that rodents can anticipate a mast (Bergeron et al., 2011; Boutin et al., 2006). Indeed, the marked positive link between red maple consumption and beech seed production strongly suggests that the systematic increased spring consumption of red maple occurred in anticipation of beech mast-seeding and not as a response to red maple production.

The cues leading to an increased consumption of red maple by chipmunks, however, remain to be determined but several hypotheses may be advanced. First, the signal may come from a secondary plant component of beech flowers, dropped to the ground after the flowering period and eaten directly upon detection (Elliott, 1978). This secondary plant component may either trigger reproduction in chipmunks (Bergeron et al., 1981), which would increase their consumption of red maple seeds to sustain the energetic costs associated with reproduction (Wade & Schneider, 1992). Secondly, the cue may come from the consumption of tissues of red maples in spring (flowers, shoots or buds) which could trigger reproduction in females and lead to a food preference for seeds once they ripen. Interestingly, there is a peak of B-vitamins in the buds and shoots of red maples in April–May (coinciding with chipmunk emergence; Elliott, 1978) followed by a drastic drop suggesting investment into seeds (Burkholder & McVeigh, 1944). We can speculate that chipmunks could get important information/essential nutrients triggering reproduction when consuming those tissues (Bergeron et al., 1981; Elliott, 1978). Alternatively, the signal may come from red maple seeds themselves, whose composition may vary between years of beech mast-seeding and post-mast years. This hypothesis would imply that red maples display a qualitative correlation, rather than a quantitative or phenological synchronization, with beeches and display different investment into seeds between beech mast and post-mast years. The seeds produced could therefore be of varying

nutritional quality depending on the year, and supposedly of higher nutritional value and thereby more appetizing in springs preceding a beech mast. This hypothesis is partly supported by Schnuur et al. (2002) who found that beech and red maple were not synchronized in their production (no. of seeds/m²) but in biomass (g of seeds/m²), suggesting that both species increased their investment in seed production in similar years.

Finally, although the best approximating model did not include any direct effects of weather on chipmunk reproduction, mild and rainy climate in June could promote summer reproduction. Indeed, some alternative candidate models suggest a potential effect of the weather in June (Figure S6, Maximum average $T_a \times$ rainfall, model C, Table S7 and Maximum T_a , model E, Table S8) that should be further investigated in the future. Moreover, weather variables could indirectly influence chipmunk reproductive success. Local weather could modulate red maple phenology, secondary plant components in flowers and buds, seed maturation or seed quality (Townsend, Wright, Beineke, Guries, & Mohn, 1982; Tremblay et al., 2002; Vasseur, Guscott, & Mudie, 2001). Considering that the month of June was rainier and cooler in beech mast than in post-mast years (Figures S6), which can influence samara production (Tremblay et al., 2002), the potential food-mediated effect of weather on chipmunk feeding ecology and reproduction should be further investigated.

5 | CONCLUSIONS

We found complex interactions between beech, red maple and chipmunks in our study system. Red maple consumption best predicts the number of females in summer oestrus, but we also found significant additional effects of red maple production. Taking altogether, our results show that red maple production per se is not a determinant of red maple consumption by chipmunks nor a reliable cue of beech mast-seeding. Results rather suggest that there is an environmental or intrinsic cue leading to an increased spring consumption of red maple by adult chipmunks in anticipation of beech mast-seeding. This increased red maple consumption appears as both a cue of the upcoming beech mast and a potential trigger of summer reproduction, which allows synchronizing juvenile emergence with the peak of beechnut availability (Bergeron et al., 2011).

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AUTHORS' CONTRIBUTIONS

M.L.T. conceived the study, and P.B., D.R. and D.G. provided inputs. Field assistants and undergraduate and graduate students collected the data. M.L.T. performed data analyses and wrote the first draft of the manuscript. P.B., D.R. and D.G. substantially participated to revisions and provided statistical guidance.

DATA AVAILABILITY STATEMENT

Data are available at Dryad Digital Repository: <https://doi.org/10.5061/dryad.stjqj2c0c> (Tissier, Réale, Garant, & Bergeron, 2020).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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